Forza di Lorentz

2/5/21 11:17 AM

a) $\vec{F} = q\,\vec{v} \times \vec{B} \Rightarrow \vec{v} \perp \vec{F}$

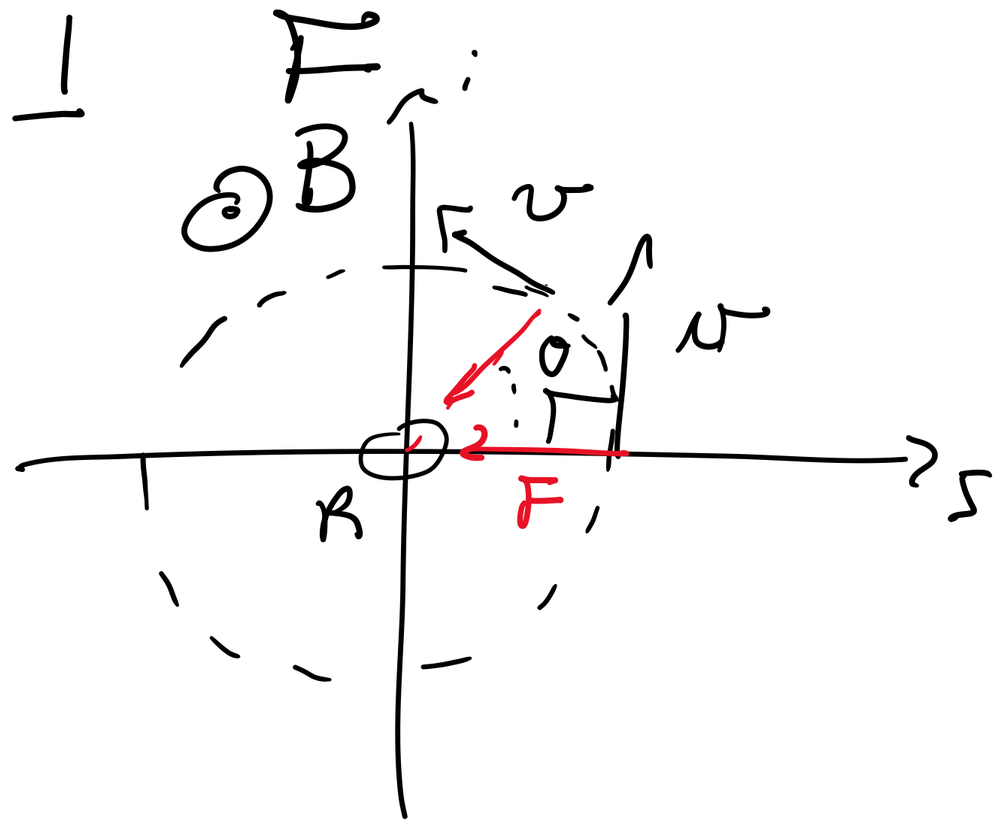
$\Delta K = \mathcal{L} \Rightarrow dK = \delta\mathcal{L}$

$\delta\mathcal{L} = \vec{F} \cdot d\vec{x} = \vec{F} \cdot \vec{v}\,dt = 0$

$\Rightarrow dK = 0 \Rightarrow \underline{K = \text{cost}}$

$\Delta \vec{p} = \int_{t_0}^{t} \vec{F}\,dt$

$v \perp F$



$\vec{F} = q\,\vec{v} \times \vec{B} \qquad \vec{v} = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$

$\vec{v} \times \vec{B} = |v||B| \begin{vmatrix} i & j & k \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{matrix} i & j \\ -\sin & \cos \\ 0 & 0 \end{matrix}$

$i\cos\theta + \sin\theta\, j$

$F = |v||B| \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$

$d\vec{p} = \vec{F}\,dt \neq 0 \Rightarrow p \neq \text{cost}$